Problem 11

Evaluate the integral:

$$\int (3^x - 4^x)\, dx.$$

Solution

To solve the given integral, we will handle each term separately. The integral can be expressed as:

$$\int (3^x - 4^x)\, dx = \int 3^x\, dx - \int 4^x\, dx.$$

Step 1: Recall the integral of an exponential function

The general formula for the integral of a^x, where $a > 0$ and $a \neq 1$, is:

$$\int a^x\, dx = \frac{a^x}{\ln(a)} + C.$$

Step 2: Apply the formula to each term

1. **First term: $\int 3^x\, dx$:** Using the formula, the integral is:

$$\int 3^x\, dx = \frac{3^x}{\ln(3)}.$$

2. **Second term: $\int 4^x\, dx$:** Similarly, using the formula, the integral is:

$$\int 4^x\, dx = \frac{4^x}{\ln(4)}.$$

Step 3: Combine the results

Substituting back into the original integral, we get:

$$\int (3^x - 4^x)\, dx = \frac{3^x}{\ln(3)} - \frac{4^x}{\ln(4)} + C,$$

where C is the constant of integration.

Final Answer

The solution to the given integral is:

$$\int (3^x - 4^x)\, dx = \frac{3^x}{\ln(3)} - \frac{4^x}{\ln(4)} + C.$$